

# EM616: Time-Series Analysis

## Introduction

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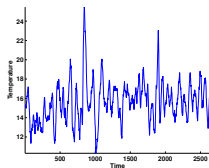
# Time-Series

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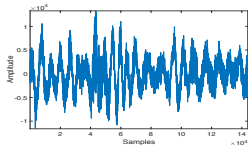
A time-series simply refers to an **ordered** collection of data (usually in time)

- ▶ e.g., yearly wages, annual production, daily temperature, hourly satellite images
- ▶ Measurements could be a function of other dimensions (e.g., frequency, space)
- ▶ Data may be collected at regular or irregular intervals
- ▶ Many variables could be recorded simultaneously (multivariate data)

# Examples



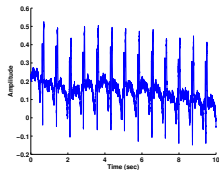
(a) Temperature



(b) Seismic noise

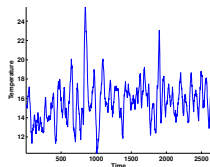


(c) Swiss SMI

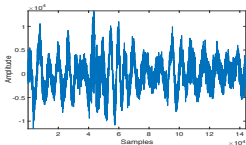


(d) ECG

# Examples



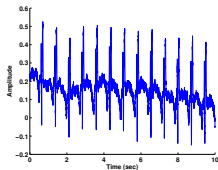
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Time-series can consist of different components:

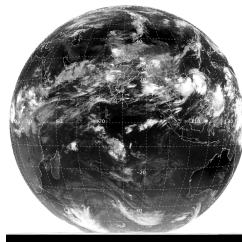
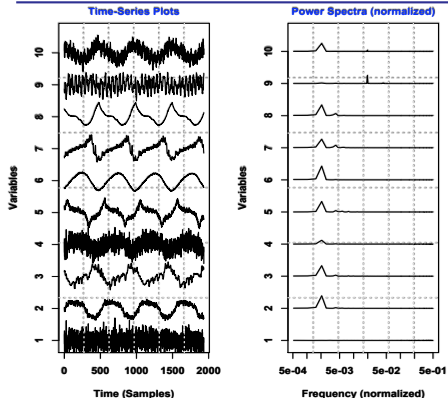
## ► Deterministic

- 1 Trend (polynomial, spline)
- 2 Seasonal (periodic) and Cyclic components

## ► Stochastic

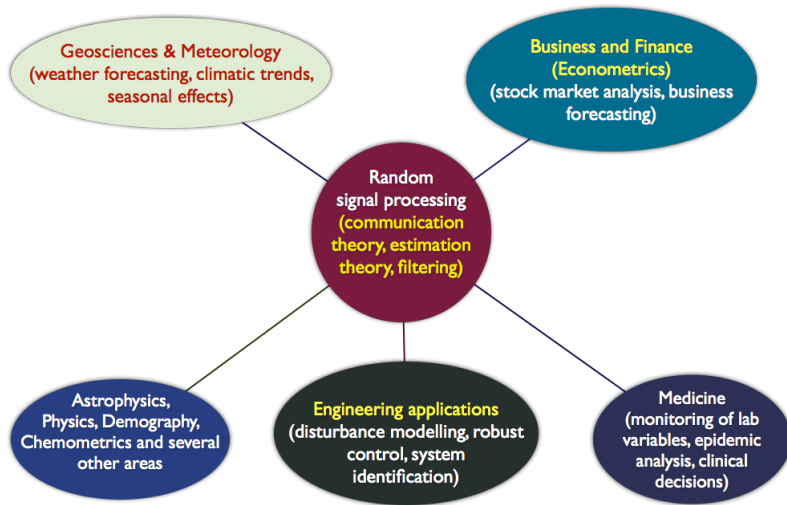
- 1 Stationary component
- 2 Non-stationary trend (e.g., random walk)
- 3 Changing mean, variance, temporal correlation, etc.

# More examples

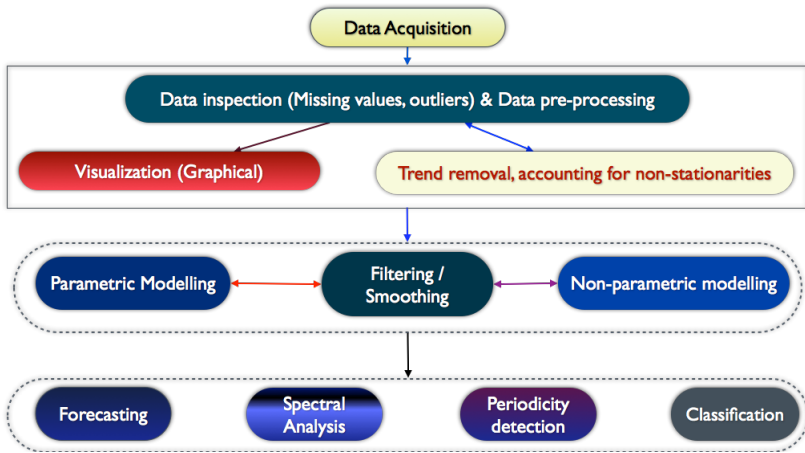


Time-series could be **multivariate**, **multidimensional** and also **transformed** into another domain

# Applications



# Systematic analysis of time-series data



# Challenges in time-series analysis

## Uncertainty (Randomness)

- i. Lack of an exact mathematical function to describe the process of interest (leading to a **probabilistic** framework).
- ii. To be able to draw inferences about the ensemble from a single realization.
- iii. Estimation of “unknowns” from (uncertain) observations / knowns.

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## Sources of uncertainty:

- 1 Process understanding (modelling errors)
- 2 Unknown causes (factors influencing the variable and disturbances)
- 3 Measurement errors

# The “realization” challenge

## Central premise

A **time-series** is an **ordered collection of random variables**, i.e., at each instant the observation is a random variable. For multivariate time-series, it is a collection of random variables.

A single time-series is said to be a **realization** of the random process.

# The “realization” challenge

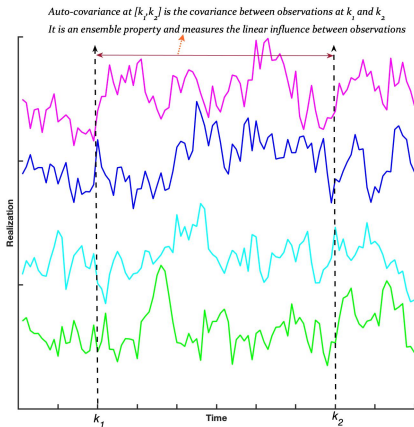
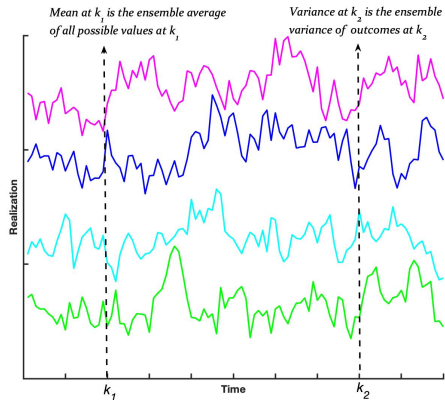
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The collection of all possible realizations is said to be the **ensemble**.

# Random Process: Ensemble of realizations



Properties of any random signal (process) are its **ensemble** properties and can change with time.

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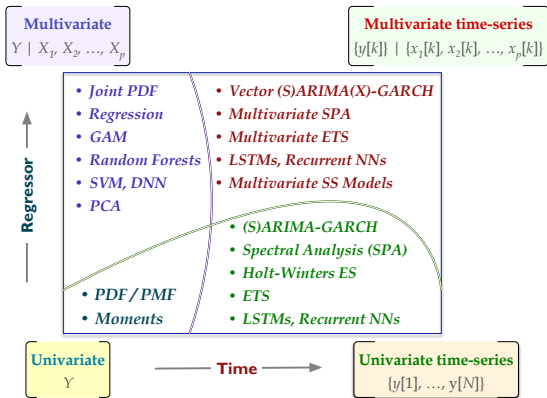
Two questions that repeatedly arise:

- 1 **How good are the estimates** (of model parameters, statistical properties, etc.) from a single realization?
- 2 **Is a single estimate sufficient?**

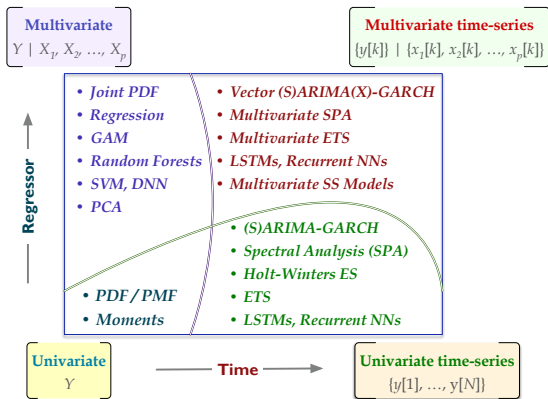
**Example:** The average reading of a variable from sensor 1 is 3.005 and from sensor 3, is 2.996. What can we say about the true level?

# Four important streets and corners

- 1 How do we predict an RV from its distribution (probability to prediction)? (rarely discussed)

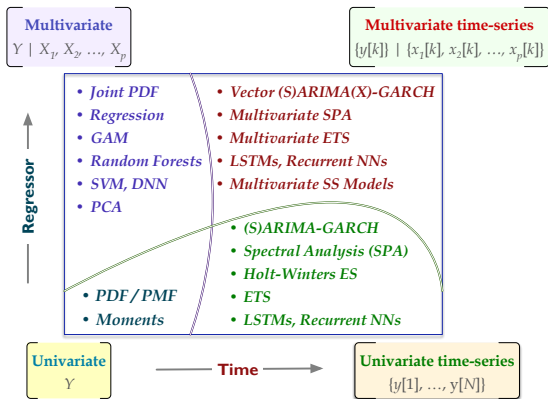


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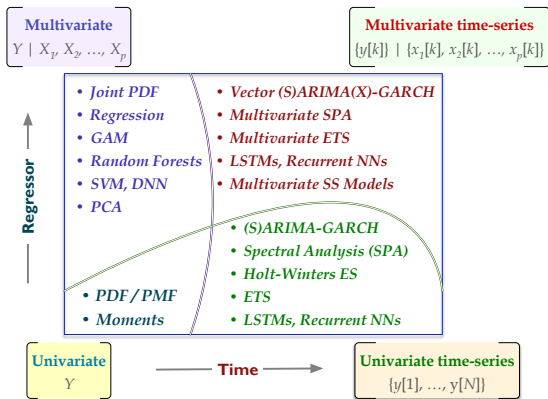
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- 2 Given the knowledge of explanatory variables, what is the best prediction? (multivariate DA)
- 3 Suppose the history (time-series) of RV is given. Does the prediction improve? If yes, how do we predict? (univariate TSA)
- 4 Finally, if both the explanatory variables and their time history are provided, what is the best prediction? (multivariate TSA)

# Estimation

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Estimators are at the heart of TSA. They produce estimates of “unobserved” or “hidden” quantities / variables from observations.

We shall learn how to:

- ▶ Characterize “goodness” of estimators
- ▶ Estimate statistical properties / parameters / signals
- ▶ Report estimation results and test hypotheses (on random processes)

# A few remarks

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- ▶ Emphasis is on **concepts** rather than on the rigour.
- ▶ Idea is to introduce and illustrate ideas first through examples.
- ▶ Student is expected to take this first-level treatment to a higher level by self-inquiry.

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- ▶ TSA requires **both theory and skill**, i.e., it is both a science and an art.
- ▶ A computational tool is indispensable to a good understanding and practice.

# R: Software for TSA

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R<sup>®</sup> is an integrated software package for data manipulation, calculation and graphical display

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- ▶ Writing user-defined and specialized add-on packages is easy
- ▶ Use of **RStudio**<sup>®</sup> makes the use of R easy and useR-friendly!

# Useful references / pre-requisites

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- ➊ **Introduction to Statistical Hypothesis Testing** as contained in the video lectures of **NPTEL / MOOC** (<http://www.nptel.ac.in/courses/103106120/>)
- ➋ Basics of linear and matrix algebra.
- ➌ **Probability Distributions (Densities) and Moments** as contained in **Lectures 01A - 09B** of **NPTEL / MOOC** video lectures on **ATSA** (<http://www.nptel.ac.in/courses/103106123/>).
- ➍ **R Tutorials for Beginners**
  - ▶ <https://youtu.be/1mQIEdhPv6I>
  - ▶ <https://youtu.be/hvXr2MHIexQ>

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